Matemàtiques
3r ESO

SOLUCIONS DE LA FITXA · UNITAT 9

Perímetres i àrees. Teorema de Pitàgores

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Si no t'ix el mateix resultat, repassa els passos o consulta els apunts interactius de la unitat: hi trobaràs una calculadora de Pitàgores i eines que calculen àrees i perímetres pas a pas. Els resultats aproximats estan arrodonits a les centèsimes amb la tecla π de la calculadora.

1 El teorema de Pitàgores

1. ★☆☆

a) $\sqrt{100} = \boxed{10 \text{ cm}}$ **b)** $\sqrt{169} = \boxed{13 \text{ cm}}$ **c)** $\sqrt{1681} = \boxed{41 \text{ m}}$

d) $\sqrt{58} \approx \boxed{7{,}62 \text{ cm}}$ **e)** $\sqrt{6{,}25} = \boxed{2{,}5 \text{ m}}$ **f)** $\sqrt{200} \approx \boxed{14{,}14 \text{ cm}}$

2. ★☆☆

a) $\sqrt{576} = \boxed{24}$ **b)** $\sqrt{225} = \boxed{15}$ **c)** $\sqrt{36} = \boxed{6}$

d) $\sqrt{119} \approx \boxed{10{,}91}$ **e)** $\sqrt{36} = \boxed{6}$ **f)** $\sqrt{65} \approx \boxed{8{,}06}$

3. ★☆☆ a) $x = \sqrt{144 + 256} = \boxed{20}$ b) $x = \sqrt{169 - 25} = \boxed{12}$ c) $x = \sqrt{289 - 225} = \boxed{8}$

4. ★★☆ L'escala és la hipotenusa: $h = \sqrt{6{,}5^2 - 2{,}5^2} = \sqrt{42{,}25 - 6{,}25} = \sqrt{36} = \boxed{6 \text{ m}}$.

5. ★★☆ $h = \sqrt{50^2 - 30^2} = \sqrt{1600} = \boxed{40 \text{ m}}$.

6. ★★☆ Els dos trams són perpendiculars: $d = \sqrt{9^2 + 12^2} = \sqrt{225} = \boxed{15 \text{ km}}$.

7. ★★☆ $\sqrt{6^2 + 0{,}8^2} = \sqrt{36{,}64} \approx \boxed{6{,}05 \text{ m}}$.

8. ★★★ $x^2 + (x+2)^2 = 100 \Rightarrow 2x^2 + 4x - 96 = 0 \Rightarrow x^2 + 2x - 48 = 0 \Rightarrow x = \frac{-2 \pm 14}{2}$: $x = 6$ ($x = -8$ no té sentit). Catets: $\boxed{6 \text{ cm i } 8 \text{ cm}}$.

9. ★★★ Si es trenca a altura x, el tros caigut mesura $9 - x$ i és la hipotenusa: $x^2 + 3^2 = (9-x)^2 \Rightarrow x^2 + 9 = 81 - 18x + x^2 \Rightarrow 18x = 72 \Rightarrow \boxed{x = 4 \text{ m}}$.

2 Classificació de triangles pels costats

10. ★☆☆

a) $169 = 169$: rectangle **b)** $64 < 85$: acutangle **c)** $81 > 52$: obtusangle

d) $676 = 676$: rectangle **e)** $144 > 113$: obtusangle **f)** $121 < 181$: acutangle

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11. ★☆☆

a) $\boxed{\text{sí}}$ **b)** no **c)** $\boxed{\text{sí}}$ **d)** $\boxed{\text{sí}}$ **e)** no

($144 + 256 = 400$; $81 + 144 = 225 \neq 256$; $225 + 1296 = 1521$; $121 + 3600 = 3721$; $64 + 81 = 145 \neq 144$.)

12. ★★☆

a) $\boxed{\text{no}}$: $9 > 3 + 5$ **b)** $\boxed{\text{sí}}$, obtusangle ($49 > 32$) **c)** $\boxed{\text{no}}$: $12 = 5 + 7$

13. ★★☆ $2{,}25 + 4 = 6{,}25 = 2{,}5^2$: $\boxed{\text{sí}}$, és rectangle. $A = \frac{1{,}5 \cdot 2}{2} = \boxed{1{,}5\ \text{m}^2}$.

14. ★★☆ $\sqrt{90^2 + 120^2} = \boxed{150\ \text{cm}}$. Amb 152 cm: $152^2 > 150^2$, l'angle oposat a la diagonal és $\boxed{\text{obtús}}$ (més de 90°).

15. ★★★ Si x és la hipotenusa: $x = \sqrt{81 + 144} = \boxed{15}$. Si la hipotenusa és 12: $x = \sqrt{144 - 81} = \sqrt{63} \approx \boxed{7{,}94}$.

16. ★★★ Comparem x^2 amb $36 + 64 = 100$, i a més ha de ser $x < 6 + 8 = 14$. Acutangle: $\boxed{x = 9}$; rectangle: $\boxed{x = 10}$; obtusangle: $\boxed{x = 11, 12, 13}$.

3 Aplicacions del teorema de Pitàgores

17. ★☆☆

a) $\sqrt{625} = \boxed{25\ \text{cm}}$ **b)** $\sqrt{98} \approx \boxed{9{,}90\ \text{cm}}$ **c)** $\sqrt{6{,}76} = \boxed{2{,}6\ \text{m}}$

18. ★☆☆ Isòsceles: $h = \sqrt{17^2 - 8^2} = \sqrt{225} = \boxed{15\ \text{cm}}$. Equilàter: $h = \sqrt{12^2 - 6^2} = \sqrt{108} \approx \boxed{10{,}39\ \text{cm}}$.

19. ★☆☆ $l = \sqrt{9^2 + 12^2} = \sqrt{225} = \boxed{15\ \text{cm}}$; $P = 4 \cdot 15 = \boxed{60\ \text{cm}}$.

20. ★★☆ $l^2 + l^2 = 144 \Rightarrow l = \sqrt{72} \approx \boxed{8{,}49\ \text{cm}}$; $P = 4\sqrt{72} \approx \boxed{33{,}94\ \text{cm}}$.

21. ★★☆ $c = \sqrt{(15 - 9)^2 + 8^2} = \sqrt{100} = \boxed{10\ \text{cm}}$; $P = 15 + 9 + 8 + 10 = \boxed{42\ \text{cm}}$.

22. ★★☆ $\frac{22 - 12}{2} = 5$; $h = \sqrt{13^2 - 5^2} = \sqrt{144} = \boxed{12\ \text{cm}}$.

23. ★★☆

a) $\sqrt{9 + 16} = \boxed{5}$ **b)** $\sqrt{36 + 64} = \boxed{10}$ **c)** $\sqrt{16 + 25} = \sqrt{41} \approx \boxed{6{,}40}$

24. ★★★ $AB = 6$, $AC = \sqrt{9 + 16} = 5$, $BC = \sqrt{9 + 16} = 5$: $\boxed{\text{isòsceles}}$, $P = \boxed{16}$. $36 < 25 + 25$: $\boxed{\text{acutangle}}$.

25. ★★★ Meitat de l'altra diagonal: $\sqrt{10^2 - 6^2} = 8$; diagonal: $\boxed{16\ \text{cm}}$.

26. ★★★ $d = 32 \cdot 2{,}54 = 81{,}28$ cm. $(16k)^2 + (9k)^2 = 81{,}28^2 \Rightarrow 337k^2 = 6606{,}44 \Rightarrow k \approx 4{,}43$. Amplària $\approx \boxed{70{,}8\ \text{cm}}$, altura $\approx \boxed{39{,}8\ \text{cm}}$.

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4 Perímetres i àrees de triangles i quadrilàters

27. ★☆☆

a) $A = \boxed{81\ \text{cm}^2}$, $P = \boxed{36\ \text{cm}}$

b) $A = \boxed{60\ \text{cm}^2}$, $P = \boxed{34\ \text{cm}}$

c) $A = \boxed{60\ \text{cm}^2}$, $P = \boxed{34\ \text{cm}}$

d) $A = \boxed{96\ \text{cm}^2}$; $l = \sqrt{64+36} = 10$, $P = \boxed{40\ \text{cm}}$

28. ★☆☆

a) $\frac{(11+7)\cdot 5}{2} = \boxed{45\ \text{m}^2}$

b) $\frac{14\cdot 9}{2} = \boxed{63\ \text{cm}^2}$

29. ★☆☆

a) $\boxed{35\,000\ \text{cm}^2}$ **b)** $\boxed{4{,}5\ \text{m}^2}$ **c)** $\boxed{23\,000\ \text{m}^2}$ **d)** $\boxed{7{,}5\ \text{m}^2}$

30. ★☆☆ $h = \frac{72}{9} = \boxed{8\ \text{cm}}$; $P = 2\cdot 9 + 2\cdot 8 = \boxed{34\ \text{cm}}$.

31. ★★☆ $h = \sqrt{64-16} = \sqrt{48} \approx 6{,}93$ cm; $A = \frac{8\cdot\sqrt{48}}{2} \approx \boxed{27{,}71\ \text{cm}^2}$.

32. ★★☆ $h = \sqrt{100-36} = 8$; $A = \frac{12\cdot 8}{2} = \boxed{48\ \text{cm}^2}$; $P = \boxed{32\ \text{cm}}$.

33. ★★☆ $h = \sqrt{13^2-5^2} = 12$; $A = \frac{(20+10)\cdot 12}{2} = \boxed{180\ \text{cm}^2}$; $P = 20+10+2\cdot 13 = \boxed{56\ \text{cm}}$.

34. ★★☆ $l = 10$; meitat de l'altra diagonal $\sqrt{100-64} = 6$, $d = 12$; $A = \frac{16\cdot 12}{2} = \boxed{96\ \text{cm}^2}$.

35. ★★☆ $A = \boxed{10\,200\ \text{m}^2} = \boxed{1{,}02\ \text{ha}}$; cost: $\boxed{30\,600\ €}$.

36. ★★★ $A = 4{,}5\cdot 2{,}6 - 1{,}2\cdot 1 = 11{,}7 - 1{,}2 = 10{,}5\ \text{m}^2$; $\frac{10{,}5}{4} = 2{,}625$: $\boxed{3\ \text{pots}}$.

37. ★★★ $A = \frac{20\cdot 15}{2} = \boxed{150\ \text{cm}^2}$; $c = \sqrt{625} = \boxed{25\ \text{cm}}$. Prenent la hipotenusa com a base: $\frac{25\cdot h}{2} = 150 \Rightarrow \boxed{h = 12\ \text{cm}}$.

38. ★★★ $b + h = 17$ i $b^2 + h^2 = 169$: $b^2 + (17-b)^2 = 169 \Rightarrow b^2 - 17b + 60 = 0 \Rightarrow b = 12$ o $b = 5$. Costats $\boxed{12\ \text{cm i } 5\ \text{cm}}$; $A = \boxed{60\ \text{cm}^2}$.

5 Polígons regulars

39. ★☆☆

a) $P = \boxed{30\ \text{cm}}$, $A = \frac{30\cdot 4{,}13}{2} = \boxed{61{,}95\ \text{cm}^2}$

b) $P = \boxed{30\ \text{cm}}$, $A = \frac{30\cdot 4{,}33}{2} = \boxed{64{,}95\ \text{cm}^2}$

c) $P = \boxed{32\ \text{cm}}$, $A = \frac{32\cdot 4{,}83}{2} = \boxed{77{,}28\ \text{cm}^2}$

40. ★☆☆ $r = l = \boxed{8\ \text{cm}}$; $a = \sqrt{64-16} = \sqrt{48} \approx \boxed{6{,}93\ \text{cm}}$; $A = \frac{48\cdot\sqrt{48}}{2} \approx \boxed{166{,}28\ \text{cm}^2}$.

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41. ★★☆ $a = \sqrt{10{,}2^2 - 6^2} = \sqrt{68{,}04} \approx \boxed{8{,}25 \text{ cm}}$; $A \approx \frac{60 \cdot 8{,}25}{2} \approx \boxed{247{,}5 \text{ cm}^2}$.

42. ★★☆ La diagonal és un diàmetre: $l^2 + l^2 = 12^2 \Rightarrow l = \sqrt{72} \approx \boxed{8{,}49 \text{ cm}}$; $a = \frac{l}{2} \approx \boxed{4{,}24 \text{ cm}}$; $A = l^2 = \boxed{72 \text{ cm}^2}$.

43. ★★☆ $a = \sqrt{100 - 25} = \sqrt{75}$; $A = \frac{60 \cdot \sqrt{75}}{2} \approx 259{,}81 \text{ cm}^2$. $\frac{30\,000}{259{,}81} \approx 115{,}5$: $\boxed{116 \text{ rajoles}}$.

44. ★★★ $a = 2$; $\frac{l}{2} = \sqrt{4^2 - 2^2} = \sqrt{12} \approx 3{,}46 \Rightarrow l \approx \boxed{6{,}93 \text{ cm}}$; $A = \frac{3 \cdot 2\sqrt{12} \cdot 2}{2} \approx \boxed{20{,}78 \text{ cm}^2}$.

6 La circumferència: longitud i arcs

45. ★☆☆

a) $14\pi \approx \boxed{43{,}98 \text{ cm}}$ **b)** $15\pi \approx \boxed{47{,}12 \text{ cm}}$ **c)** $5\pi \approx \boxed{15{,}71 \text{ m}}$

46. ★☆☆ $r = \frac{62{,}83}{2\pi} \approx \boxed{10 \text{ cm}}$; $d \approx \boxed{20 \text{ cm}}$.

47. ★☆☆

a) $4\pi \approx \boxed{12{,}57 \text{ cm}}$ **b)** $2\pi \approx \boxed{6{,}28 \text{ cm}}$ **c)** $5\pi \approx \boxed{15{,}71 \text{ cm}}$

48. ★★☆ Cada volta: $60\pi \approx 188{,}50$ cm. $\frac{150\,000}{60\pi} \approx \boxed{795{,}8 \text{ voltes}}$ (unes 796).

49. ★★☆ Les dues semicircumferències fan una circumferència: $200 + 2\pi \cdot 36{,}5 = 200 + 73\pi \approx \boxed{429{,}34 \text{ m}}$.

50. ★★☆ 20 min són $\frac{1}{3}$ de volta (120°): $\frac{2\pi \cdot 15}{3} = 10\pi \approx \boxed{31{,}42 \text{ cm}}$.

51. ★★★ $\frac{2\pi \cdot 12 \cdot n}{360} = 18{,}85 \Rightarrow n = \frac{18{,}85 \cdot 360}{24\pi} \approx \boxed{90°}$.

52. ★★★ $L = 2\pi \cdot 150\,000\,000 \approx 942\,477\,796$ km; 1 any $= 365 \cdot 24 = 8760$ h; $v = \frac{L}{8760} \approx \boxed{107\,589 \text{ km/h}}$.

7 Àrea del cercle i figures circulars

53. ★☆☆

a) $25\pi \approx \boxed{78{,}54 \text{ cm}^2}$ **b)** $36\pi \approx \boxed{113{,}10 \text{ cm}^2}$ **c)** $0{,}64\pi \approx \boxed{2{,}01 \text{ m}^2}$

54. ★☆☆

a) $6\pi \approx \boxed{18{,}85 \text{ cm}^2}$ **b)** $\frac{125\pi}{3} \approx \boxed{130{,}90 \text{ cm}^2}$ **c)** $3\pi \approx \boxed{9{,}42 \text{ cm}^2}$

55. ★☆☆ $\pi(81 - 16) = 65\pi \approx \boxed{204{,}20 \text{ cm}^2}$.

56. ★★☆ $r = \sqrt{\frac{200}{\pi}} \approx \boxed{7{,}98 \text{ cm}}$; $L = 2\pi r \approx \boxed{50{,}13 \text{ cm}}$.

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57. ★★☆ $\frac{\pi(144-49)\cdot 80}{360} = \frac{190\pi}{9} \approx \boxed{66{,}32\ \text{cm}^2}$.

58. ★★☆ $A = \frac{225\pi}{8} \approx \boxed{88{,}36\ \text{cm}^2}$; arc: $\frac{30\pi}{8} \approx \boxed{11{,}78\ \text{cm}}$.

59. ★★☆ $l = r = 5$: circumscrit $25\pi \approx \boxed{78{,}54\ \text{cm}^2}$. $a^2 = 25 - 6{,}25 = 18{,}75$: inscrit $18{,}75\pi \approx \boxed{58{,}90\ \text{cm}^2}$.

60. ★★★ Sector: $\frac{36\pi\cdot 60}{360} = 6\pi$. Triangle equilàter de costat 6: $h = \sqrt{27}$, $A = 3\sqrt{27} \approx 15{,}59$. Segment: $6\pi - 3\sqrt{27} \approx \boxed{3{,}26\ \text{cm}^2}$.

61. ★★★ $\pi(100 - r^2) = 50\pi \Rightarrow r^2 = 50 \Rightarrow r = \sqrt{50} \approx \boxed{7{,}07\ \text{cm}}$.

8 Figures compostes i problemes

62. ★☆☆ $A = 6\cdot 6 + \frac{6\cdot 4}{2} = 36 + 12 = \boxed{48\ \text{m}^2}$. Costats del teulat: $\sqrt{3^2+4^2} = 5$; $P = 6+6+6+5+5 = \boxed{28\ \text{m}}$.

63. ★☆☆ $A = 64 + \frac{\pi\cdot 4^2}{2} = 64 + 8\pi \approx \boxed{89{,}13\ \text{cm}^2}$; $P = 3\cdot 8 + \frac{2\pi\cdot 4}{2} = 24 + 4\pi \approx \boxed{36{,}57\ \text{cm}}$.

64. ★★☆ Els dos semicercles fan un cercle de radi 3: $A = 60 - 9\pi \approx \boxed{31{,}73\ \text{cm}^2}$; $P = 2\cdot 10 + 2\pi\cdot 3 = 20 + 6\pi \approx \boxed{38{,}85\ \text{cm}}$.

65. ★★☆ $A = 144 - \frac{\pi\cdot 144}{4} = 144 - 36\pi \approx \boxed{30{,}90\ \text{cm}^2}$.

66. ★★☆ $A = 1{,}4\cdot 1{,}8 + \frac{\pi\cdot 0{,}7^2}{2} = 2{,}52 + 0{,}245\pi \approx \boxed{3{,}29\ \text{m}^2}$; $P = 1{,}4 + 2\cdot 1{,}8 + 0{,}7\pi \approx \boxed{7{,}20\ \text{m}}$.

67. ★★☆ Rectangle exterior: $(25+4)\times(12+4) = 29\cdot 16 = 464$; $A = 464 - 300 = \boxed{164\ \text{m}^2}$.

68. ★★★ Mitja fulla = quart de cercle − mig quadrat $= 25\pi - 50$. $A = 2(25\pi - 50) = 50\pi - 100 \approx \boxed{57{,}08\ \text{cm}^2}$.

69. ★★★ Radi 5: $A = 300 - 3\cdot 25\pi = 300 - 75\pi \approx \boxed{64{,}38\ \text{cm}^2}$.

70. ★★★ $r = 4$, $a = \sqrt{16-4} = \sqrt{12}$; hexàgon: $\frac{24\sqrt{12}}{2} = 12\sqrt{12} \approx 41{,}57$; $A = 16\pi - 12\sqrt{12} \approx 50{,}27 - 41{,}57 = \boxed{8{,}70\ \text{cm}^2}$.

71. ★★★ Tres quarts de cercle de radi 5 i, en doblegar les cantonades, dos quarts de cercle de radi $5-4=1$: $A = \frac{3}{4}\cdot 25\pi + 2\cdot\frac{\pi}{4} = 19{,}25\pi \approx \boxed{60{,}48\ \text{m}^2}$.

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